University of Engineering & Technology Peshawar

Programming Fundamentals (C++) Lab

LAB REPORT 4

STUDENT NAME	**MUHAMMAD SAHAL AYUB**
ETEA ID:	**4042**
COUSRE:	**EE-170LCOMPUTER PROGRAMMING LAB**
DATE:	**3/2/26**
LAB TITLE:	**SWITCH STATEMENT**

SUBMITTED TO ENGR RIZWAN ULLAH

Learning Objectives:

- Learn and practice switch statements in C++.
- Implement menu-driven programs and case-based decision making.
- Understand how break statements prevent fall-through in switch.
- Apply switch statements to solve real-world programming problems

Theory:

Switch Statement:

A switch statement allows a variable to be tested for equality against a list of values (cases).

Syntax:

```
switch(expression) {

    case value1:

        // statements

        break;

    case value2:

        // statements

        break;

    ...

    default:

        // statements if no case matches

}
```

Key points:

- Each **case** ends with a break statement to prevent fall-through.
- **default** executes if no cases match.
- **switch** works with integers, characters, or enums, but not with floats or strings.

Example:

```
char grade = 'B';

switch(grade) {

  case 'A': cout << "Excellent"; break;
```

```
    case 'B': cout << "Very Good"; break;
    default: cout << "Invalid grade";
}
```

Lab Tasks & Programs

Task 1: Day name using switch statement

Description:

Program takes a number (1–7) and prints the corresponding day name.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int day;
    cout << "Enter day number (1-7): ";
    cin >> day;

    switch(day) {
        case 1: cout << "Monday"; break;
        case 2: cout << "Tuesday"; break;
        case 3: cout << "Wednesday"; break;
        case 4: cout << "Thursday"; break;
        case 5: cout << "Friday"; break;
        case 6: cout << "Saturday"; break;
        case 7: cout << "Sunday"; break;
```

```
        default: cout << "Invalid day number";
    }

    cout << endl;
    return 0;
}
```

Output:

```
Enter day number (1-7): 3
Wednesday
```

Task 2:Grade Evaluation Using Switch Statement

Description: Program takes a character grade (A–F) and prints evaluation.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    char grade;  // Character type variable

    cout << "Enter your grade (A, B, C, D, F): ";
    cin >> grade;

    switch(grade)  // Grade ki value check karega
    {
```

```
        case 'A':
            cout << "Excellent";
            break;
        case 'B':
            cout << "Very Good";
            break;
        case 'C':
            cout << "Good";
            break;
        case 'D':
            cout << "Pass";
            break;
        case 'F':
            cout << "Fail";
            break;

        default:  // Agar koi aur letter ho
            cout << "Invalid Grade";
    }

    return 0;
}
```

Output:

```
 Enter your grade (A, B, C, D, F): B
Very Good
```

Task 3: Unit Conversion Method Using Switch Statement

Description: Program shows a menu for conversions and performs the selected operation.

Code:

```
#include <iostream>
using namespace std;


int main()
{
    int choice;      // Menu choice store karega
    double value;    // Input value store karega
    double result;   // Result store karega

    cout << "1 -> Km to Meters" << endl;
    cout << "2 -> Meters to Cm" << endl;
    cout << "3 -> Kg to Grams" << endl;
    cout << "4 -> Celsius to Fahrenheit" << endl;

    cout << "Enter your choice: ";
    cin >> choice;

    switch(choice)  // User ki choice check karega
    {
        case 1:
            cout << "Enter Kilometers: ";
```

```
            cin >> value;
            result = value * 1000;  // Km ko meters mein convert
            cout << "Meters = " << result;
            break;

        case 2:
            cout << "Enter Meters: ";
            cin >> value;
            result = value * 100;   // Meters ko cm mein convert
            cout << "Centimeters = " << result;
            break;

        case 3:
            cout << "Enter Kilograms: ";
            cin >> value;
            result = value * 1000;  // Kg ko grams mein convert
            cout << "Grams = " << result;
            break;

        case 4:
            cout << "Enter Celsius: ";
            cin >> value;
            result = (value * 9/5) + 32;  // Celsius ko Fahrenheit mein convert
            cout << "Fahrenheit = " << result;
            break;
```

```
        default:
            cout << "Invalid choice";
    }

    return 0;
}
```

Output:

```
 Conversion Menu:
1 → Kilometers to Meters
2 → Meters to Centimeters
3 → Kilograms to Grams
4 → Celsius to Fahrenheit
Enter your choice: 4
Enter Celsius: 25
25 °C = 77 °F
```

Conclusion:

- Learned how to use switch statements for multiple case-based decisions.
- Understood the importance of break to avoid fall-through.
- Applied switch statements to practical problems like day names, grade evaluation, and unit conversions.
- Learned to create menu-driven programs in C++ for better user interaction.